Lok Jagruti Kendra University
University with a Difference

wef: 2025-26

L.J. INSTITUTES OF COMPUTER APPLICATIONS			
Programme	Master of Computer Applications	**Branch/Spec.**	Computer Applications
Semester	II	**Effective from Academic Year**	2025-26
Subject Category	Major	**Effective for the batch Admitted in**	June 2025
Subject Code	240110202	**Subject Name**	Data Structure Lab (DS-Lab)

Teaching scheme						**Examination scheme (Marks)**			
(Lecture hours Per week)	**Credit**	**L**	**T**	**P**	**Total**	**E**	**CEC**	**V**	**Total**
4	2	0	0	4	48	0	50	50	100

University Exam Evaluation Scheme

Internal Evaluation		**External Evaluation**	
Mid Sem (50 Marks)	**30**	**Theory Exam**	**NA**
Attendance	**10**	**Practical Exam**	**50**
Hacker Rank Practical Assignment	**10**		
Total Marks	**50**	**Total Marks**	**50**

Pre-requisites:

- Basic knowledge of writing and understanding algorithms for solving a problem.
- Knowledge of programming in C language.

Program Learning Outcome:

Name of PO	**Description**
PO1	Foundational & Theoretical Mastery
PO2	Software Design & Development Expertise
PO3	Problem-Solving & Analytical Ability
PO4	Proficiency in Emerging Technologies
PO5	Information Systems & Management Perspective
PO6	Professional & Ethical Responsibility
PO7	Communication & Teamwork
PO8	Lifelong Learning & Career Readiness

Course Learning Outcome:

Name of CO	**Description**
CO1	To understand how to solve a problem step by step to get the desired output with different way.

CO2	To learn the optimal way to organize data in the digital space.
CO3	To understand the fundamental concept of elementary data structures and their implementation.
CO4	To understand where and how the data structures are implemented to solve the real world problems.
CO5	To learn how to write efficient and optimized computer programs.

Mapping of CO and PO:

Course Learning Objectives	Program Outcomes (POs)							
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**
CO1	3	2	2	-	-	-	-	1
CO2	2	3	3	-	-	-	-	1
CO3	2	3	3	-	2	2	-	1
CO4	2	3	3	2	-	2	-	1
CO5	1	3	2	2	-	-	2	1

Content:

Unit	Content	Weightage (%)	No of Hrs.	CO
Unit I	1. Write a program to reverse a number. 2. Write a program to find sum of digits 3. Write a program to display array elements 4. Write a program to find sum and average of array elements 5. Write a program to find largest and smallest element in array 6. Write a program to access array elements using pointers. 7. Write a program to allocate memory using malloc() 8. Write a program to allocate memory using calloc()	**10%**	**6**	**CO1**
Unit II	1. Write a program to find the total of each rows and columns in a given matrix. 2. Program to reverse an array and store it in another array. 3. Write a menu driven program to perform following operations for an array a) Print b) Insert at a specific position c) Delete at a specific position d) Update at specific position e) Exit 4. Write a program for linear search in array 5. Write a program using singly linked list to perform following operations: a) Create b) Display c) Insert - First, Last, At a specified location d) Delete -First, Last, At a specified location 6. Perform the following operations on singly linked list using user-defined functions: a) Count total nodes b) Create a copy of the linked list c) Merge	**25%**	**12**	**CO2**

	7. Perform the following operations on singly linked list using user-defined functions: a) Reverse a listed list b) Search for a specific node c) Sort 8. Write a program to count the number of occurrences of a user specified element in the singly linked list. If that element does not present give an appropriate message 9. Write a program to display all the nodes of a singly linked list which is divisible by 7. If not found, then display appropriate message. 10. Write a program to display all the even node of a singly linked list. If not found, then display appropriate message 11. Write a program to search an existing element in a singly linked list. If not found, then display appropriate message 12. Implement Stack using linked list 13. Implement simple queue using linked list 14. Write a program using doubly linked list to perform following operations a) Create b) Display c) Insert (At all locations) d) Delete (At all locations) e) Count f) Exit 15. Write a program using doubly linked list to perform following operations a) Copy b) Search 16. Write a program using doubly linked list to perform merge operation			
Unit III	1. Write a menu driven program to perform following operations in stack using static data structure. Also display the updated value of Top in every operation. a) Push b) Pop c) Peep d) Count e) Display f) Exit 2. Write a program to reverse a given number using stack. 3. Write a program to convert infix expression to postfix 4. Write a program to evaluate arithmetic expression from postfix expression. 5. Write a program to check balanced parenthesis in expression using stack 6. Write a menu driven program to perform following operations in simple queue using static data structure. Also display the updated value of Front and Rear in every operation. a) Insert b) Delete c) Display d) Exit 7. Write a menu driven program to perform following operations in circular queue using static data structure. Also display the updated value of Front and Rear in every operation. a) Insert b) Delete c) Display d) Exit	**25%**	**12**	**CO3**
Unit IV	1. Write program to create and display binary tree 2. Write program to create and display binary search tree 3. Write a program to display tree traversals In-order traversal (recursive)	**20%**	**10**	**CO4**

	Pre-order traversal (recursive) Post-order traversal (recursive) 4. Write a program to create a graph using adjacency matrix 5. Write a program to display degree of each vertex in a graph			
Unit V	1. Write a program for following sorting techniques. a) Bubble b) Selection c) Insertion d) Shell e) Merge f) Quick 2. Write a program implement binary search algorithm	**20%**	**8**	**CO5**

Reference Books:	
1	Debasis Samanta, Classic Data Structures , PHI, Second Edition.
2	Jean-Paul Tremblay, Paul G. Sorenson, "An Introduction to Data Structures with Applications", Tata McGraw-Hill, 2nd Edition, (2007).
3	Mark Allen Weiss, “Data Structures and Algorithm Analysis in C”, Pearson, Second Edition,
4	Ashok N. Kamthane, “Introduction to Data Structures in C”, Pearson Education (2004).
5	Reema Thareja, Data Structures using C, Oxford
Web Reference:	
1	**https://www.tutorialspoint.com/data_structures_algorithms**
2	**https://www.javatpoint.com/**
3	**GeeksforGeeks (Data Structures & Algorithms):** https://www.geeksforgeeks.org
Question Paper Scheme:	
	University Examination Duration: 2 Hours 30 minutes. **SECTION-I** Q-1: **(Unit I & II)** (15 Marks) Q-2: **(Units III & IV)**(15 Marks) Q-3: **(Unit V)**. (10 Marks) **SECTION-II** VIVA (10 Marks)